Introduction to Deep Equilibrium Nets (DEQNs)

Mini-Course · March 2026

Galo Nuño (Banco de España) · Simon Scheidegger (HEC, University of Lausanne)

FOUNDATIONS & METHODOLOGY

Part I — ML / DL Foundations

DNNs as Function Approximators
Activation Functions & Layers
Stochastic Gradient Descent
Loss Functions & Training

Slides: Part I

Part II — Deep Equilibrium Nets

Motivation & Curse of Dimensionality
DEQN Loss Function Design
Training Algorithm
Economic Equilibrium Conditions

Slides: Part II

Part III — Brock-Mirman Benchmark

Analytical Solution for Validation
DEQN Implementation
Stochastic Extensions (TFP)
Parallelization & Scalability

Notebooks 01 – 03

TOOLS, SCALING & PRACTICE

Part IV — NAS & ReLoBRaLo

Neural Architecture Search
Hyperparameter Tuning
Multi-Task Loss Balancing
ReLoBRaLo Normalization

Slides: Part IV

Part V — Scaling Up: IRBC Model

Multi-Country Dynamics
Equilibrium System & Complementarity
Gauss–Hermite Quadrature
High-Dimensional DEQNs

Notebook 04

Part VI — Hands-On Exercises

Coding Exercises (Fill-in)
Model Implementation
Suggested Readings
Individual Discussions

Notebook 03 (Exercises)

Theory · Economic Methods · Computational Tools · Practice

Tools: Python · TensorFlow · TF Probability · PyTorch · NumPy · SciPy · Matplotlib